Questions by topic: Dynamic equilibria, feedback & tipping points - Set 1 (60 marks)

Answer all questions in the spaces provided.

0 1 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

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0 1 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

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0 1 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

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0 1 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

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0 2 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 2 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

0 2 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 3 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 3 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 3 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 4 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 4 . 1 Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO_2 levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

0 4 . 2 Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

0 4 . 3 Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

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0 4 . 4 Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

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0 5 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

0 5 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

0 5 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

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0 5 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

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0 6 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

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0 6 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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0 6 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

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0 6 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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Extra space

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END OF QUESTIONS